

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED

COURSE CODE: CSA15

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Code of Conduct

Total Marks: 25

I S.SUBASRI (192511296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Application-Based Questions – Cloud Computing

Below are professional, exam-ready answers for all 5 questions. Each answer includes a real-world scenario, step-by-step explanation, architecture flow, advantages, and a suitable picture.

1. Auto-Scaling and Load Balancing for Seasonal E-Commerce Traffic

AWS E-Commerce Refer



admin.cloudairy.com



Elastic Load
Balancing ELB

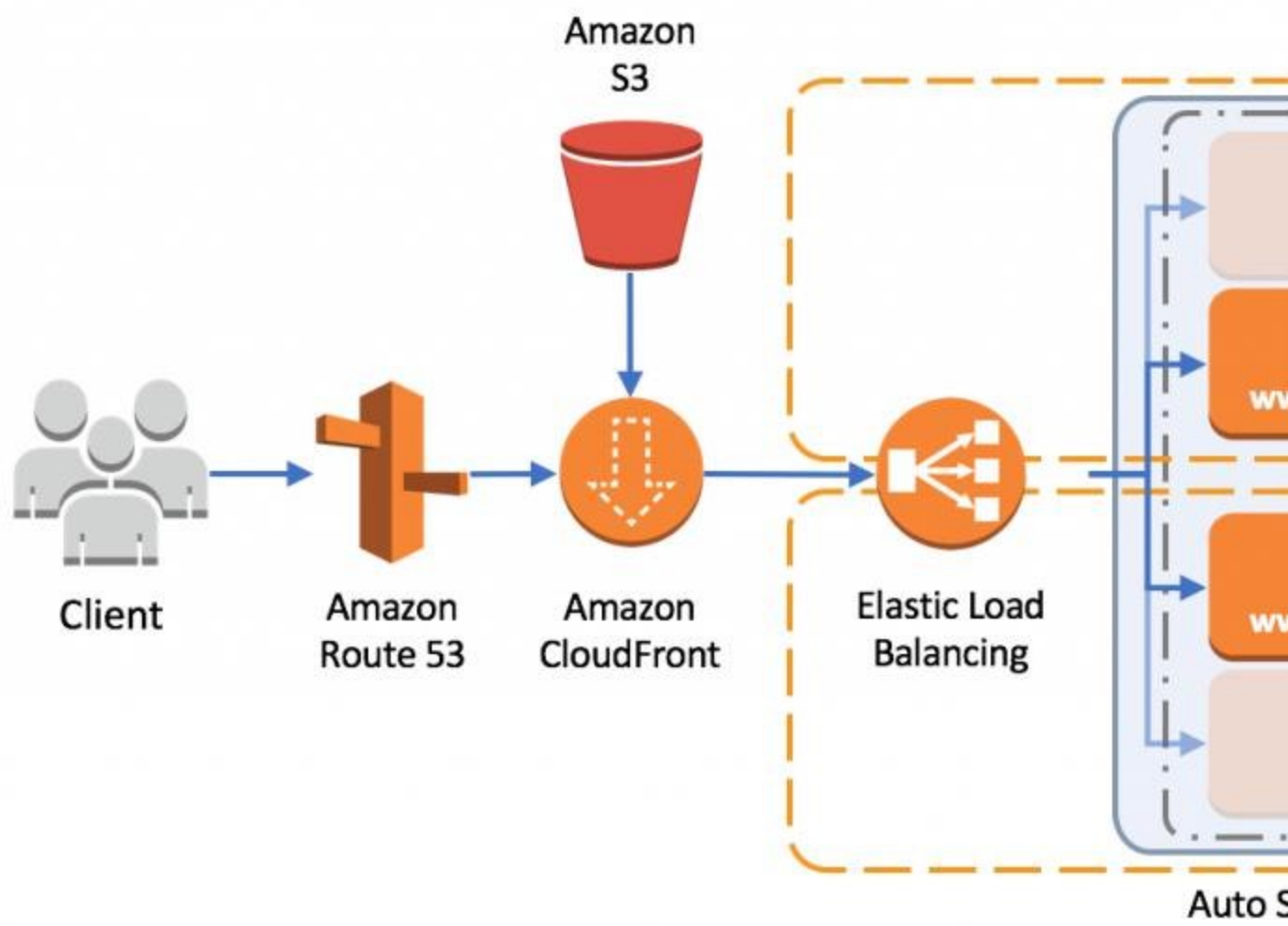
3



EC2 Auto
Scaling



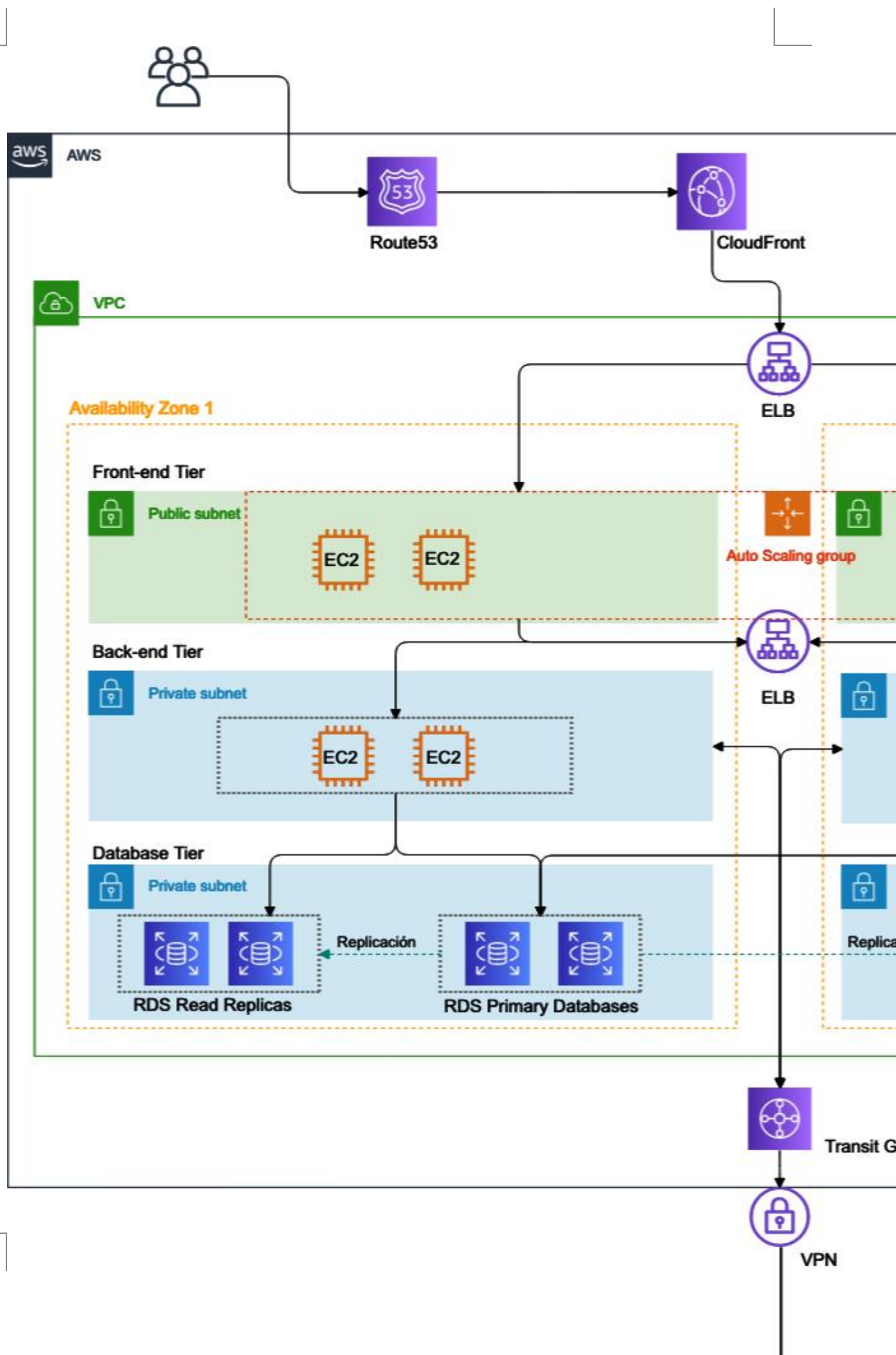
Analys



aws Auto Scaling



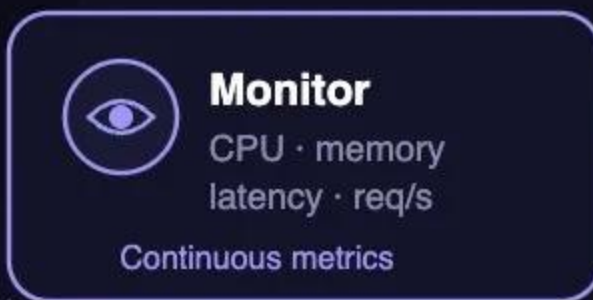
AUTOMATIC ELASTICITY



How Auto Scal

Automatically adjusts resources in re

DEMAND OVER TIME



Continuous loop — re-evaluates every cycle

Scale-Out

High demand →

Add new instances automatically



+



2 instances

4 instances

Trigger: CPU > 80% for 5 min

→ +2 spun up

Auto scaling ensures optimal performance and co

Introduction

During seasonal events such as Black Friday, Diwali sales, Christmas sales, or flash sales, an e-commerce website may suddenly receive thousands or millions of requests. If the company uses a fixed number of servers, the servers may become overloaded, resulting in slow response times, errors, or downtime.

Cloud auto-scaling and load balancing work together to solve this problem.

How Auto-Scaling Works

Auto-scaling automatically increases or decreases computing resources according to demand.

During normal traffic:

Users Load Balancer 2 Servers

During traffic spike:

Users Load Balancer 2 5 10 Servers

When CPU utilization, request count, or response latency increases, the cloud platform can automatically add instances. When demand decreases, unnecessary instances can be removed. ([AlibabaCloud](#))

Role of Monitoring Tools

Monitoring tools continuously collect:

- CPU and memory utilization
- Number of users
- Network traffic
- Request rate
- Response time
- Error rate
- Server health

For example, a monitoring service can detect:

CPU usage > 70% Trigger Auto Scaling Add new servers.

Monitoring therefore acts as the feedback mechanism for auto-scaling.

How It Reduces Downtime

1. Traffic increases.
2. Monitoring detects increased workload.
3. Auto-scaling launches additional servers.
4. Load balancer distributes requests.
5. Failed servers are removed from traffic.
6. Users continue accessing the website.

Real-World Example

Consider an Amazon/Flipkart-style e-commerce platform during a Diwali sale.

Normally:

10,000 users 4 servers

During the sale:

500,000 users Auto Scaling 20 servers

The load balancer distributes customers across these servers. After the sale, the system scales down again.

Benefits

Feature	Benefit
Auto-scaling	Handles sudden traffic increases
Load balancing	Prevents individual servers from becoming overloaded
Monitoring	Detects performance problems
Health checks	Removes failed servers

Multi-zone
deployment

Improves availability

Conclusion: Auto-scaling provides the required capacity, while load balancing distributes traffic efficiently. Together they provide high availability, lower downtime, better performance, and improved customer experience.

2. Secure Cloud Storage for Backup Data

ENTERPRISE BACKUP

SECURE DATA
AGAINS

APPLICATION SERVERS



BACKUP SOFTWARE



CLOUD

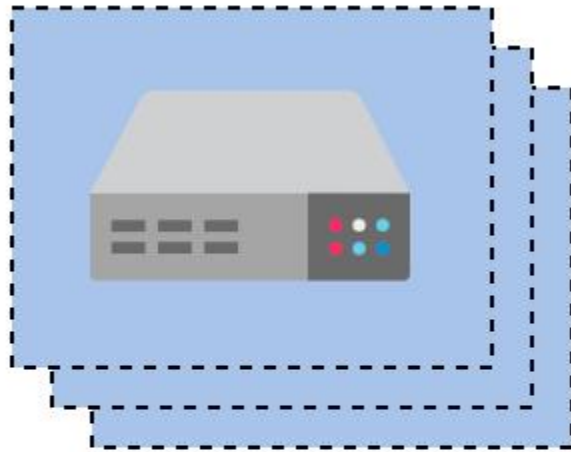


ENCRYPTED



IMMUTABLE

Zone 2



Restored app servers



Restored databases

Restore

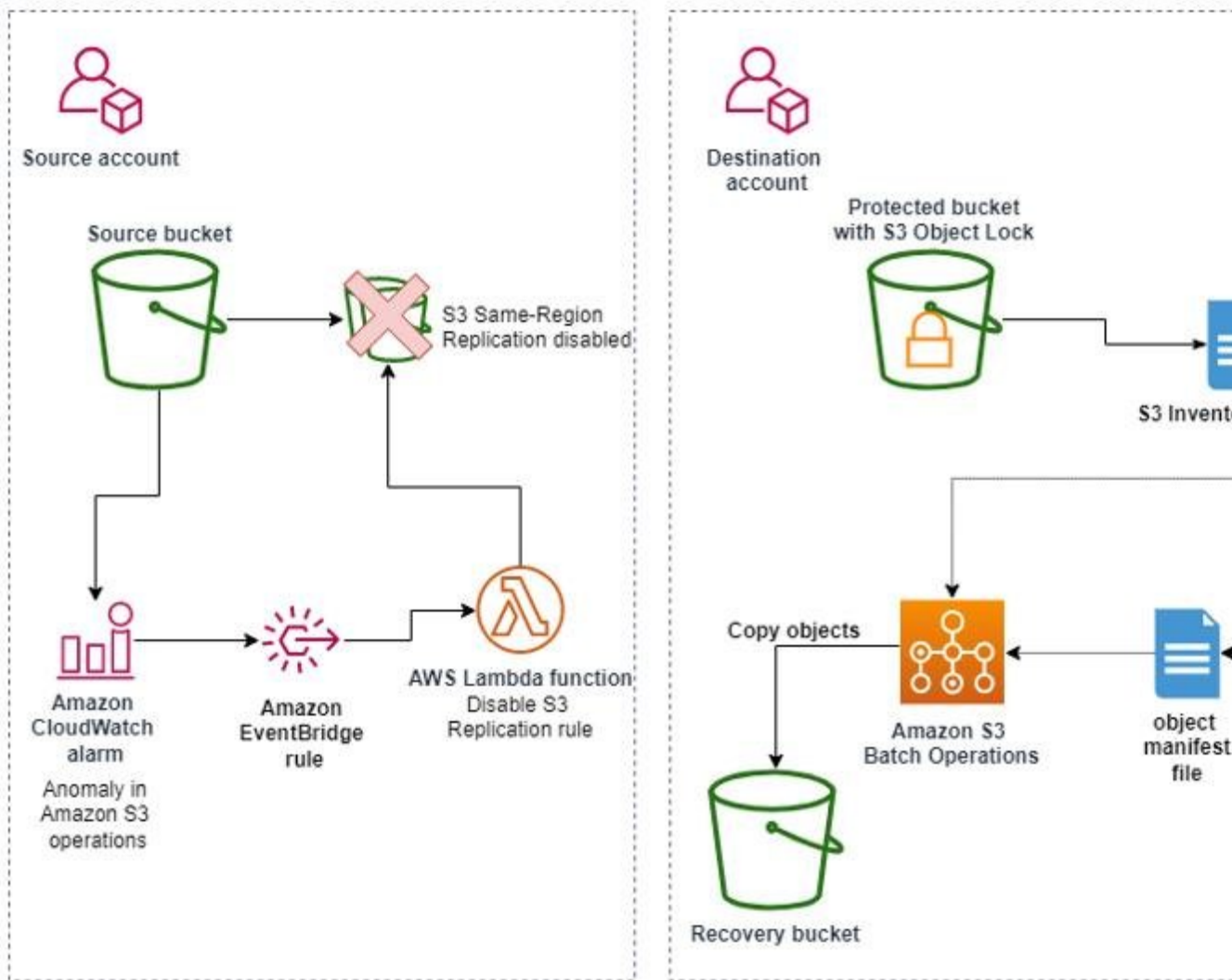


S3 cross-region
replication

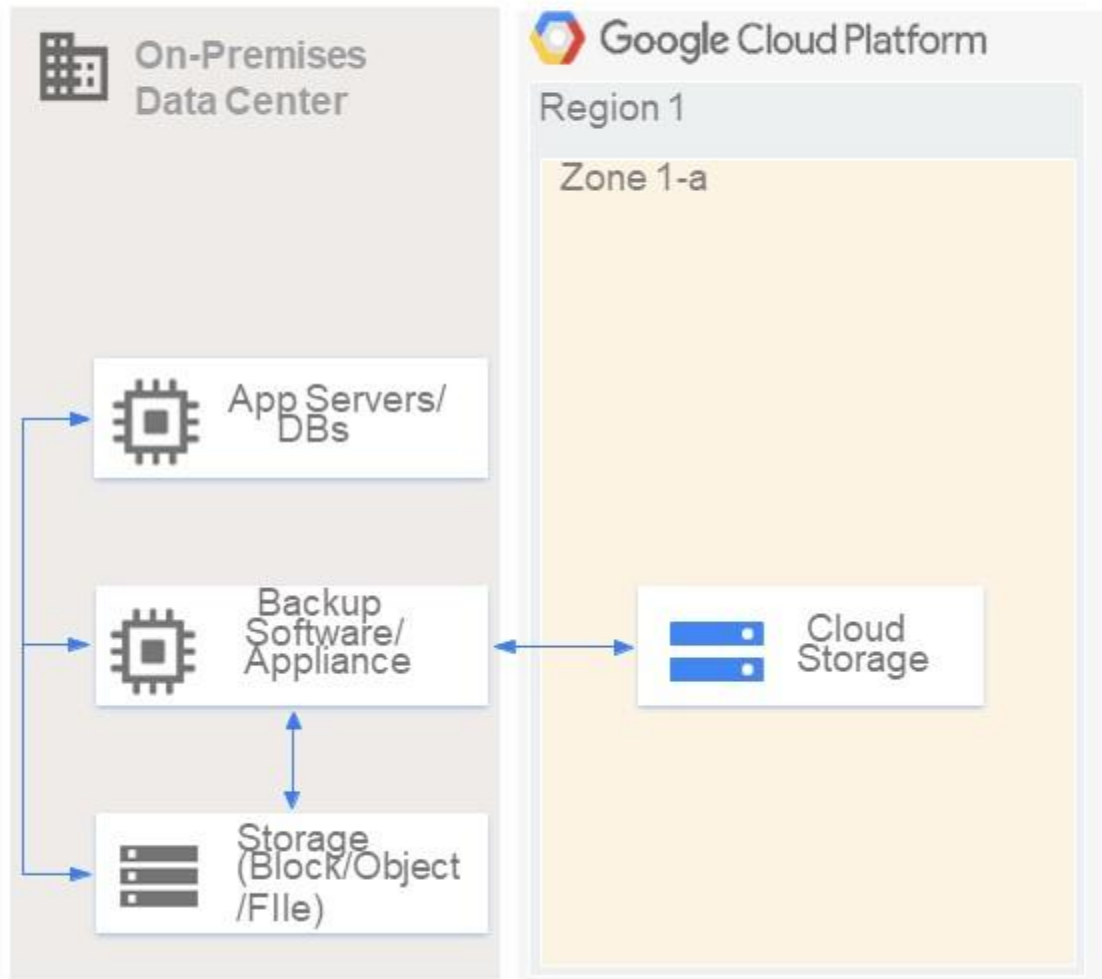


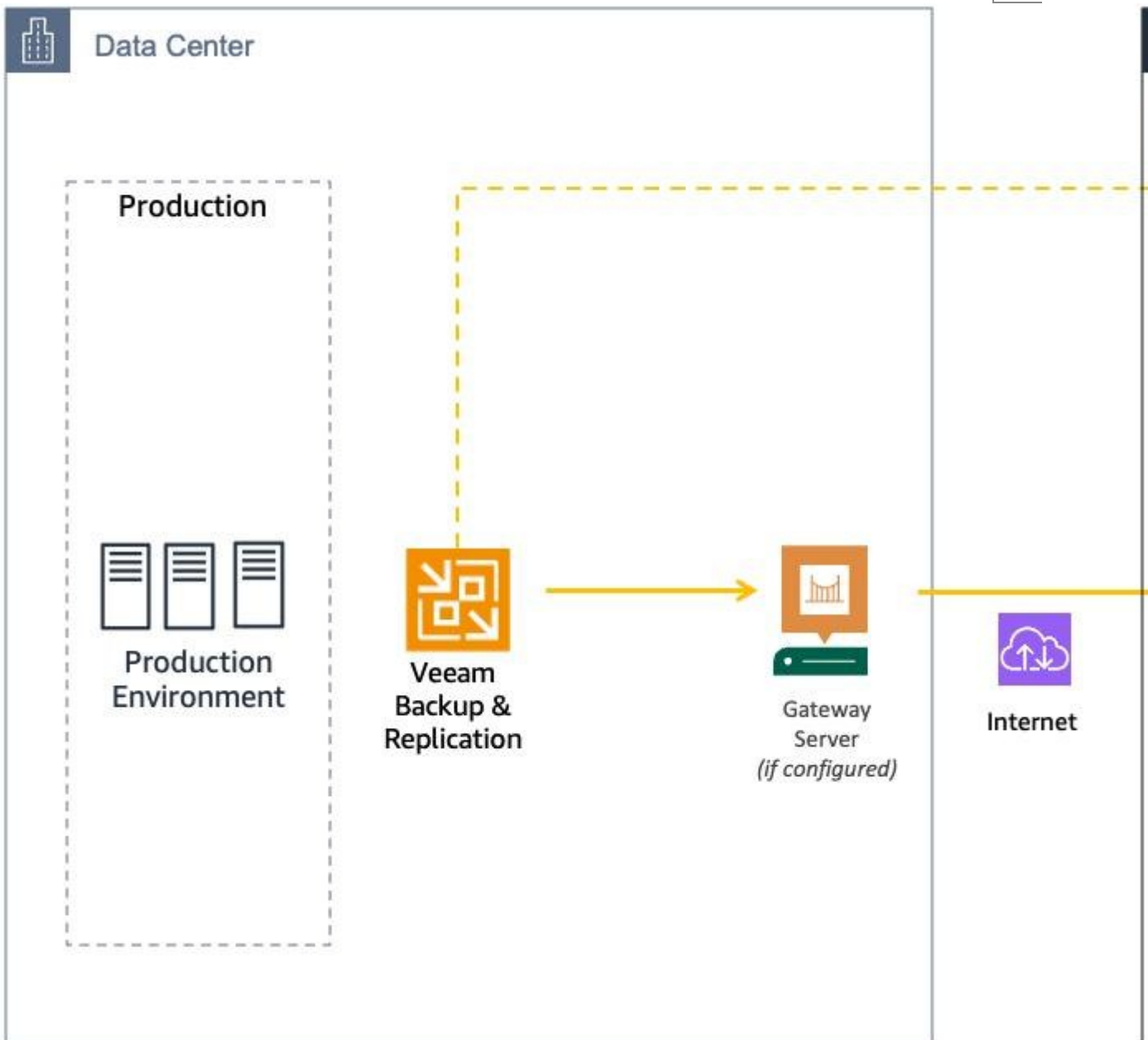


us-east-1



Backup to cloud

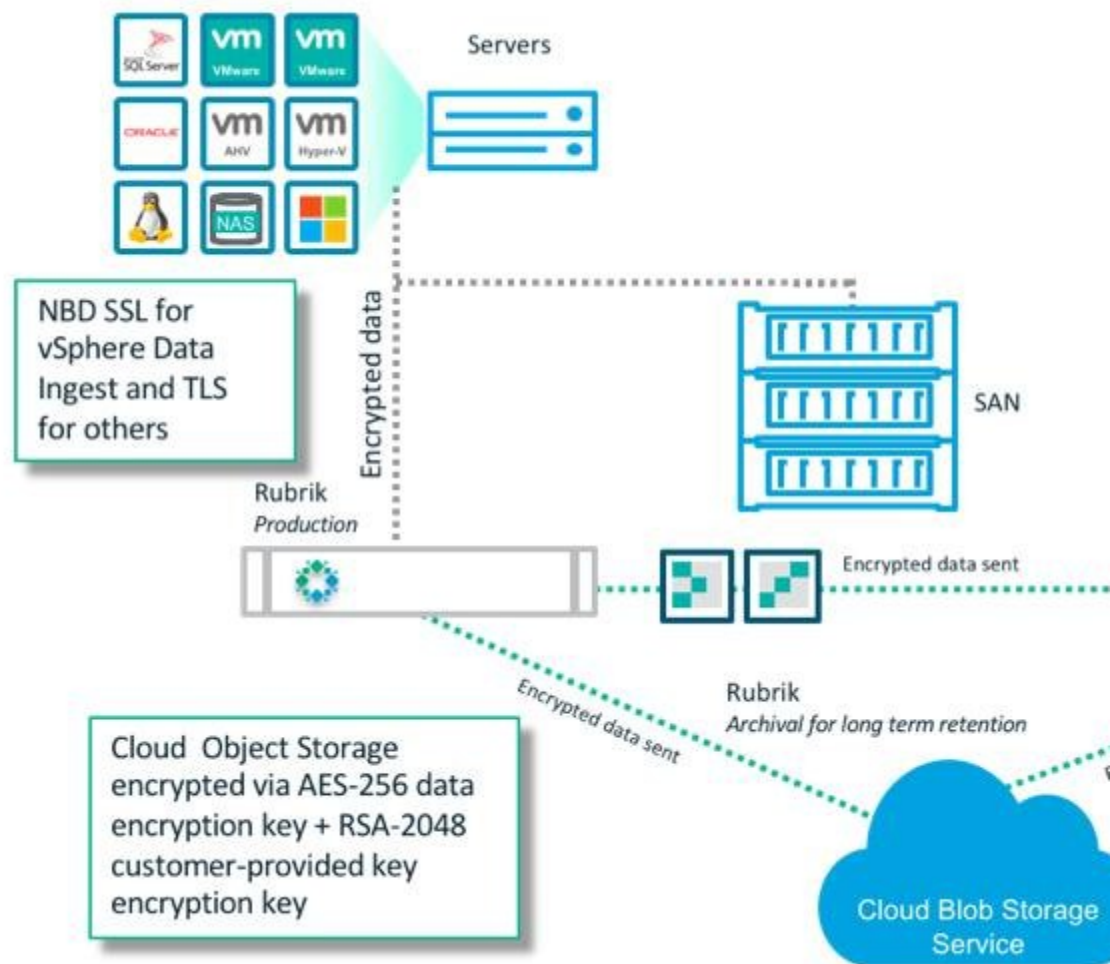




Geo-Resilient Cloud Architecture



Multi-Region Failover & Disaster Recovery



Introduction

Organizations generate huge amounts of backup data such as:

- Database backups
- Customer records
- Application files
- Documents
- Transaction history
- System images

Cloud storage provides durability, redundancy, encryption, access control, and cost-effective storage.

1. Data Durability

Cloud storage providers store data redundantly across physical infrastructure.

For example, Amazon S3 is designed for 99.999999999% durability, and cloud backup services can maintain multiple copies of data. ([AWS Documentation](#))

2. Data Redundancy

Redundancy means maintaining multiple copies of data.

BACKUP DATA

Cloud Storage

Copy 1 Copy 2 Copy 3

Region A Region A Region B

If hardware or even an entire location becomes unavailable, another copy can be used.

Cloud storage can also replicate objects across regions for greater geographic protection. ([AWS Documentation](#))

3. Encryption

Encryption converts readable data into protected ciphertext.

At rest:

Backup Encryption Cloud Storage

In transit:

Application TLS/HTTPS Cloud Storage

Cloud storage services support encryption at rest and secure transmission using TLS. ([AWS Documentation](#))

4. Access Control

Access control ensures that only authorized users can access backup data.

Examples include:

- Identity and Access Management (IAM)
- Role-based access control
- Multi-factor authentication
- Bucket policies
- Private storage
- Least-privilege permissions

For example:

Administrator	Full Backup Access
Backup Service	Write Access
Developer	No Backup Access
Auditor	Read/Audit Access

5. Cost Advantages

Traditional backup systems may require:

- Physical servers
- Hard disks
- Tape libraries
- Data-center space
- Cooling
- Electricity

- Hardware maintenance
- IT staff

Cloud storage uses a pay-as-you-use model and provides different storage classes for frequently and infrequently accessed data. ([Amazon Web Services, Inc.](#))

Traditional Storage	Cloud Storage
Buy hardware	Pay for required capacity
Data-center space required	Provider manages infrastructure
Manual scaling	Automatic scalability
Hardware maintenance	Provider-managed
Expensive expansion	Expand on demand
Limited geographic redundancy	Multi-region options

6. Disaster Recovery Example

Suppose a company's primary data center is destroyed by a fire or flood.

Normal architecture:

Primary Data Center

Continuous Backup

Cloud Backup Storage

Disaster Recovery Site

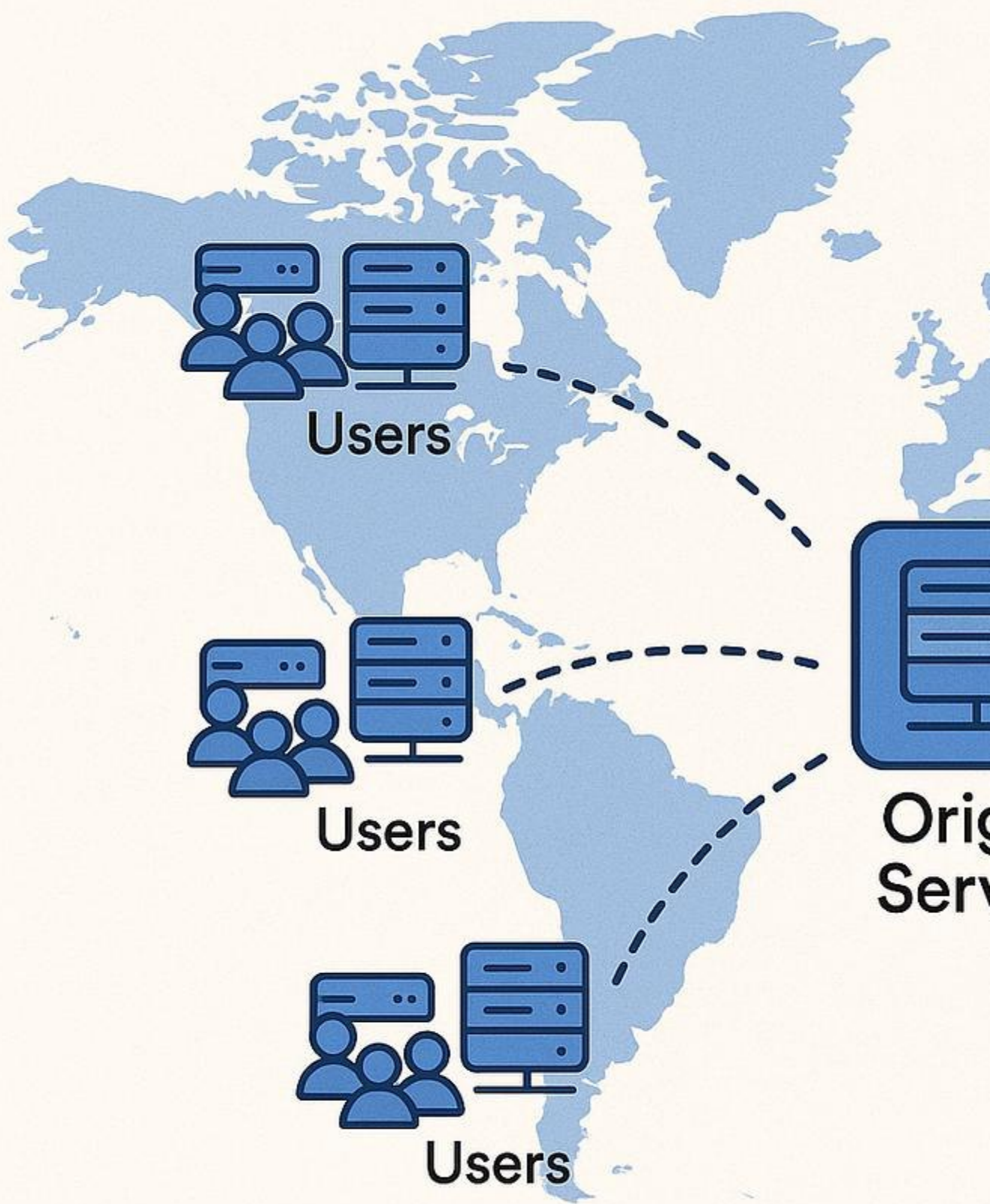
After the disaster:

1. Primary servers become unavailable.
2. Company launches recovery infrastructure.
3. Latest cloud backup is retrieved.
4. Databases and applications are restored.
5. Business operations resume.

Cloud backup services can also use versioning and object-locking mechanisms to protect against accidental deletion or modification. ([AWS Documentation](#))

Conclusion: Cloud storage provides high durability, redundancy, encryption, access control, scalability, and lower infrastructure costs, making it highly suitable for enterprise backup and disaster recovery.

3. CDN and Global Deployment of a Mobile Application





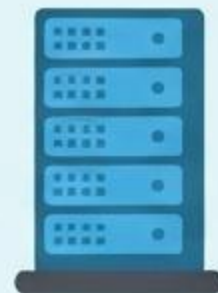
**Video Source &
Encoder**



 **Live Content**



On-Demand Content



Origin Server

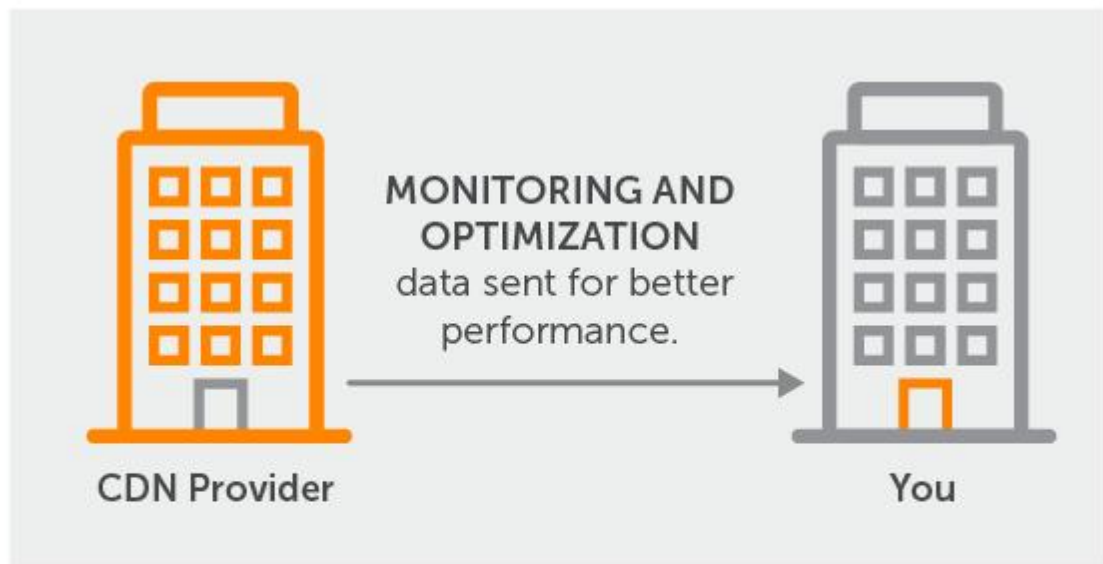
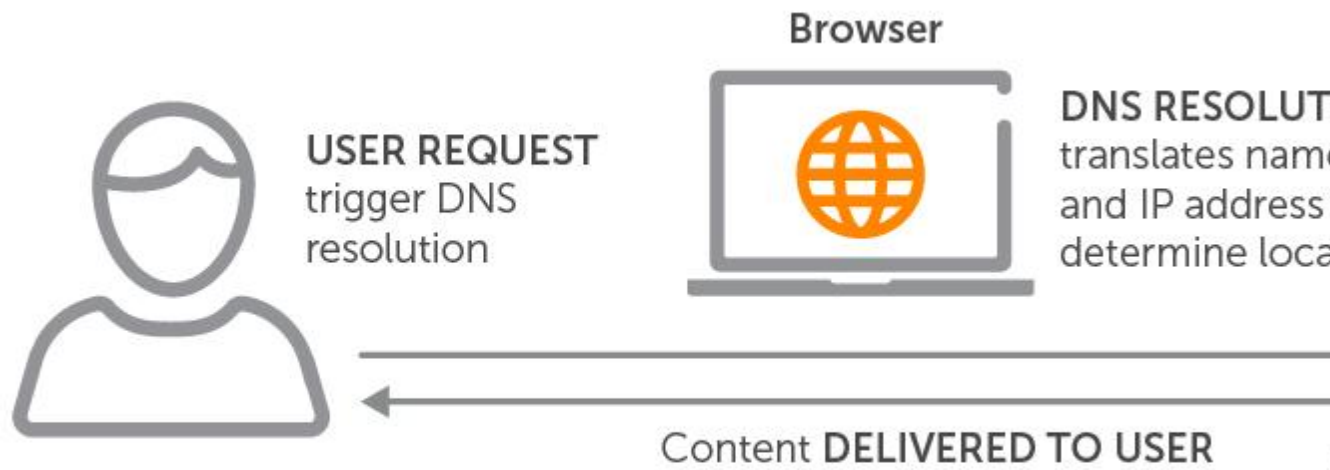


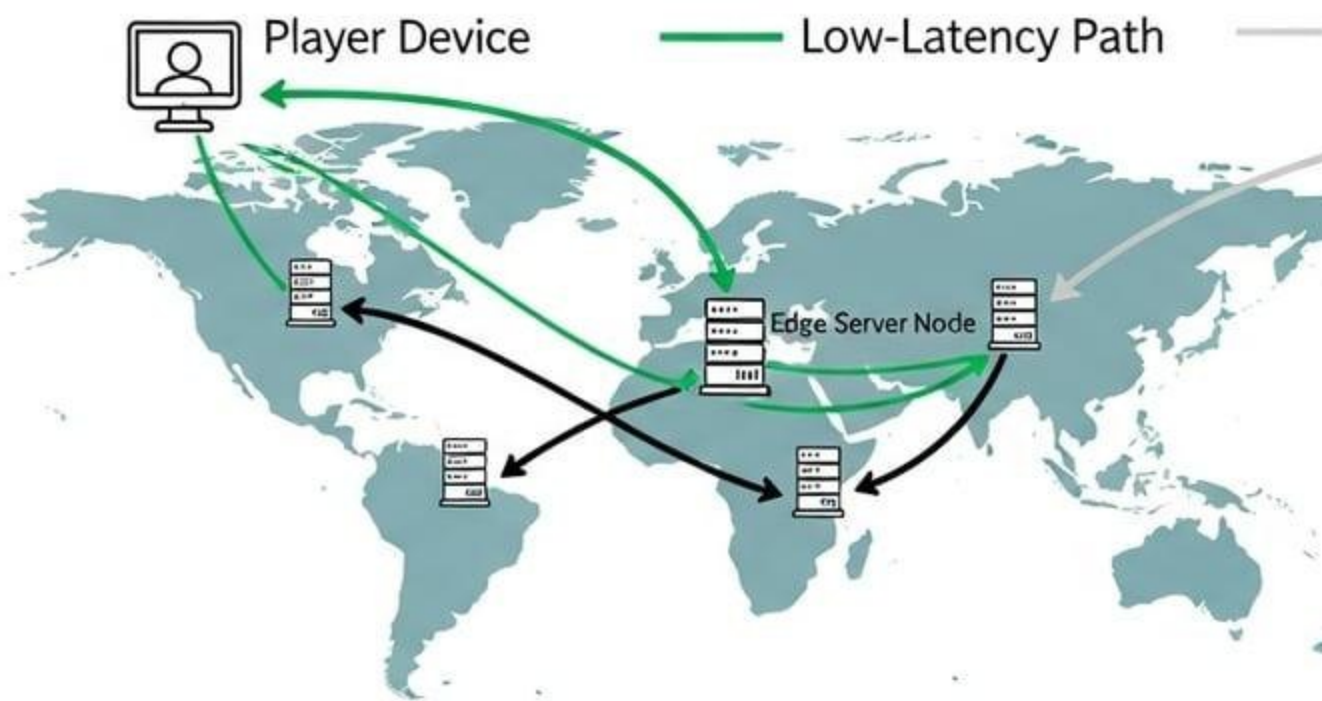
Edge

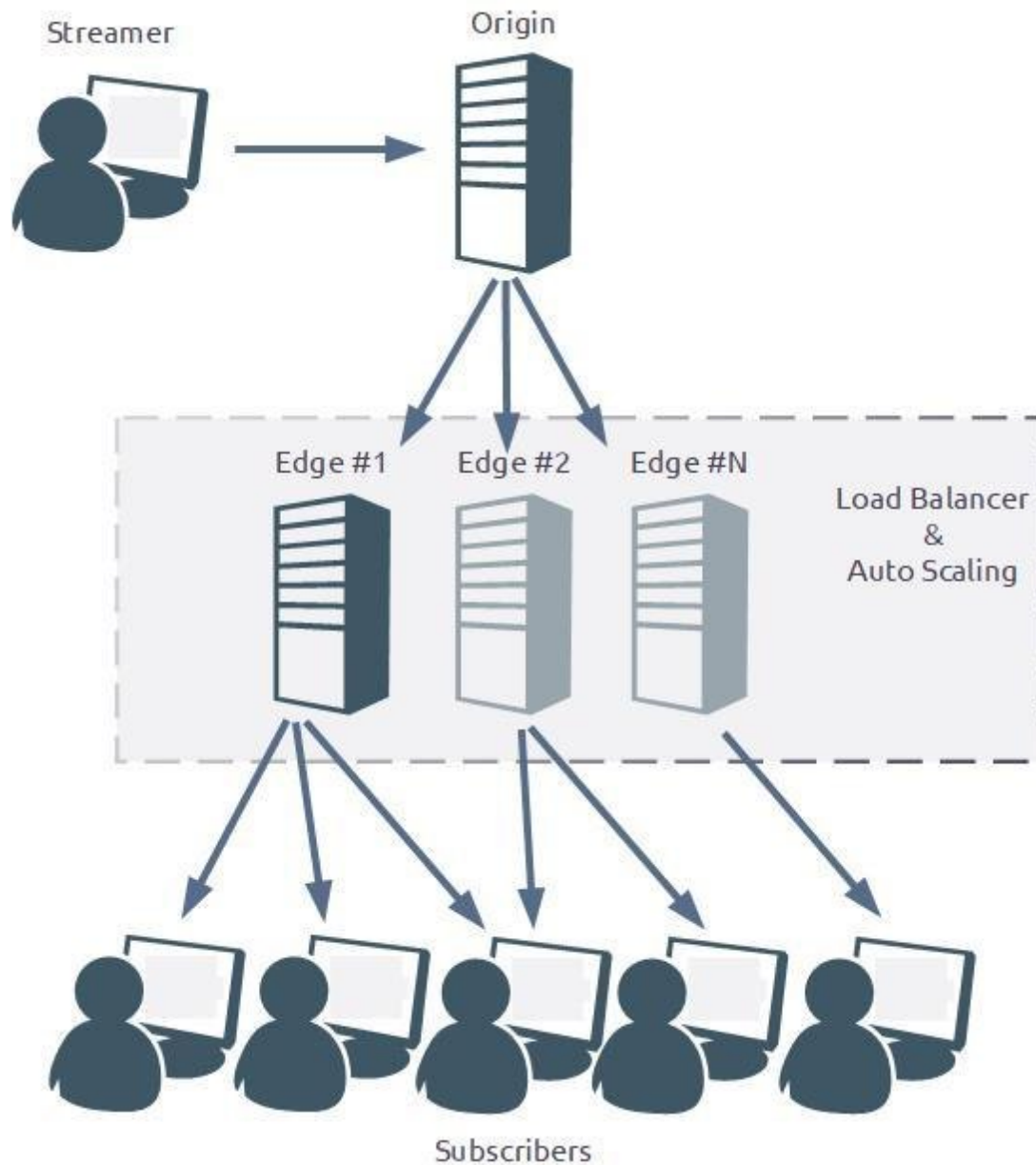
Cach

How a CDN Helps C Deliver Large Gam









Introduction

When a mobile application is deployed globally, users may be located thousands of kilometers away from the main application server.

For example:

India

Europe

USA

Japan

Origin Server

Long distances can increase network latency.

A Content Delivery Network (CDN) solves this by placing content closer to users.

How CDN Works

A CDN consists of many geographically distributed edge servers.

ORIGIN SERVER

Edge Asia Edge Europe Edge USA

Users Users Users

The CDN caches frequently accessed content such as:

- Images
- Videos
- JavaScript
- CSS
- Game assets
- App downloads
- Static API responses

Instead of every request going to the origin server, users can receive cached content from a nearby edge location. ([Cloudflare Docs](#))

How Edge Servers Reduce Latency

Suppose the origin server is located in the USA.

Without CDN:

Indian User USA Server Indian User

With CDN:

Indian User India/nearby Edge Server Indian User

Therefore:

Shorter network distance Lower latency Faster response

CDNs are designed to place cached content close to end users and improve performance and availability. ([Cloudflare Docs](#))

Global Availability

Major cloud providers support:

- Multiple geographic regions
- Availability zones
- Global load balancing
- CDN/edge locations
- Traffic routing
- Replication
- Disaster recovery

This allows applications to serve users from infrastructure closer to them.

Gaming Example

For an online multiplayer game:

- Game assets CDN
- Updates CDN
- User authentication Cloud backend
- Game servers Regional cloud regions
- Players Nearest game region

This reduces download time and can improve responsiveness.

Benefits

CDN Feature	Advantage
Edge caching	Faster content delivery
Geographic distribution	Lower latency
Reduced origin traffic	Less server load
Global infrastructure	Better availability
Edge security	Protection against some network attacks
Caching	Reduced bandwidth consumption

Cloudflare, for example, describes CDN edge caching as a way to reduce latency by serving content from locations closer to users. ([Cloudflare](#))

Conclusion: CDN and edge computing bring content and services closer to users, resulting in lower latency, faster loading, improved availability, and a better global user experience.

4. Cloud Compliance and Data Security for a Financial Institution

MODERN CLOUD SECURITY ARCH

LAYER 1: NETWORK SECURITY



**Public
Internet**



**Web Application
Firewall & DDoS
Protection**



LAYER 2: DATA ENCRYPTION



Encryption at Rest

**RDS/S3
Encrypted Storage**

- AES-256 Encryption
- AWS KMS / Azure Key Vault
- Auto Key Rotation



Bank XYZ



**AWS
Organisations**



**Business
Unit 1**



Cloud Trail



**Business
Unit 2**



Cloud Trail



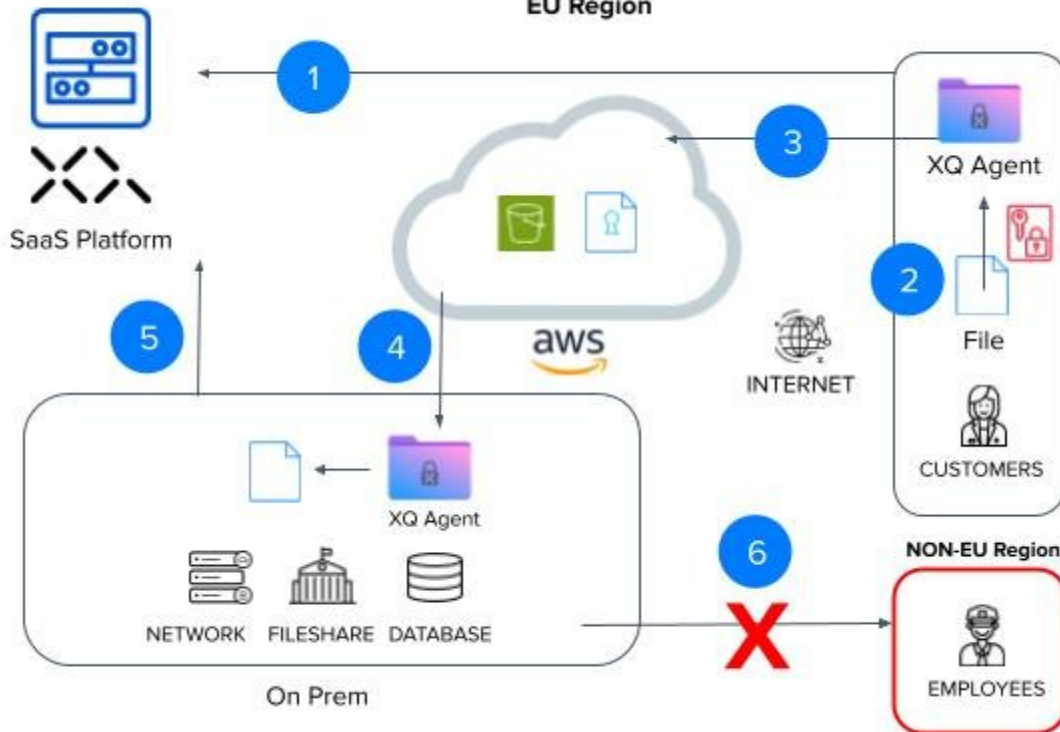
**Business
Unit 3**



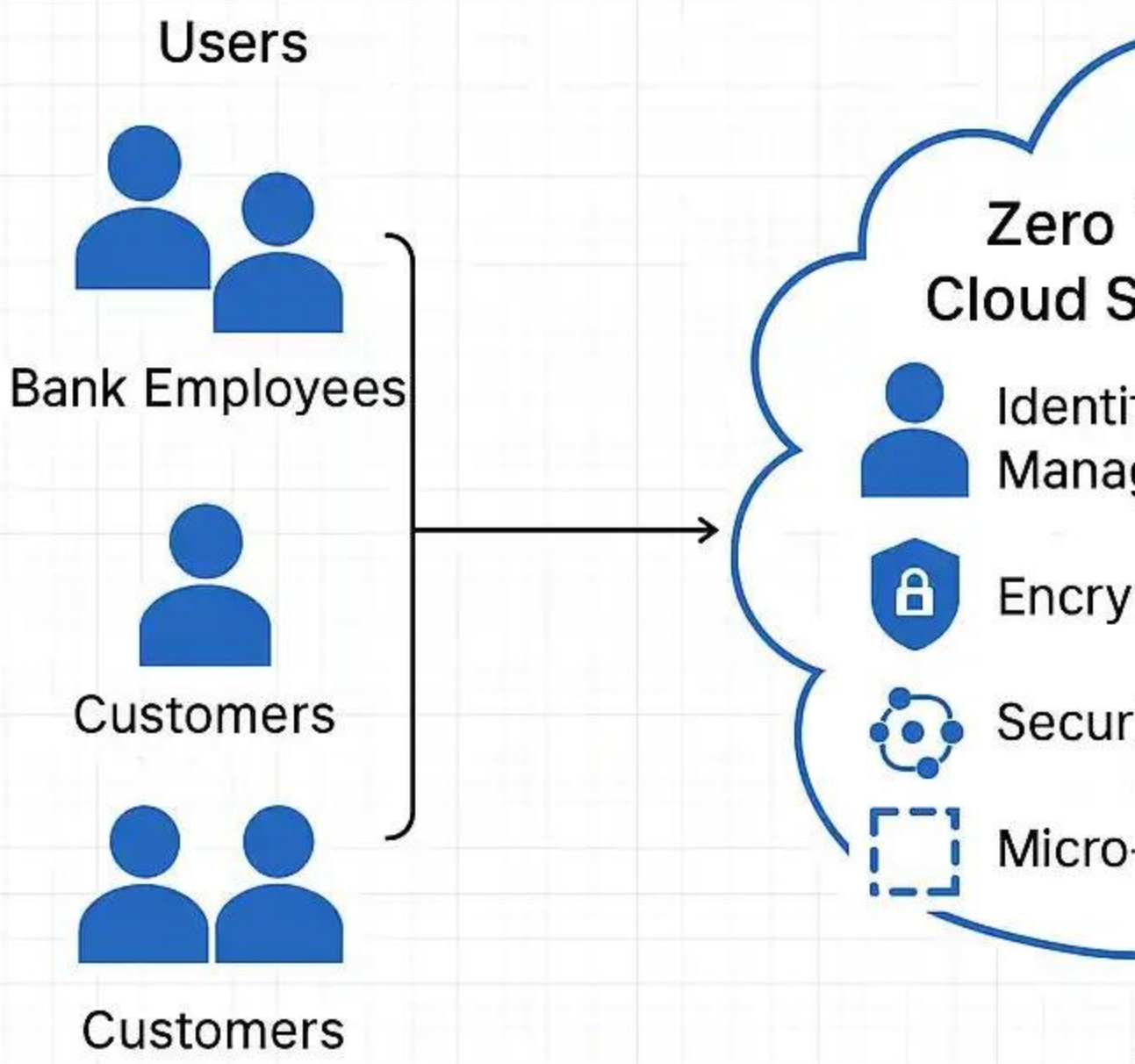
Cloud Trail

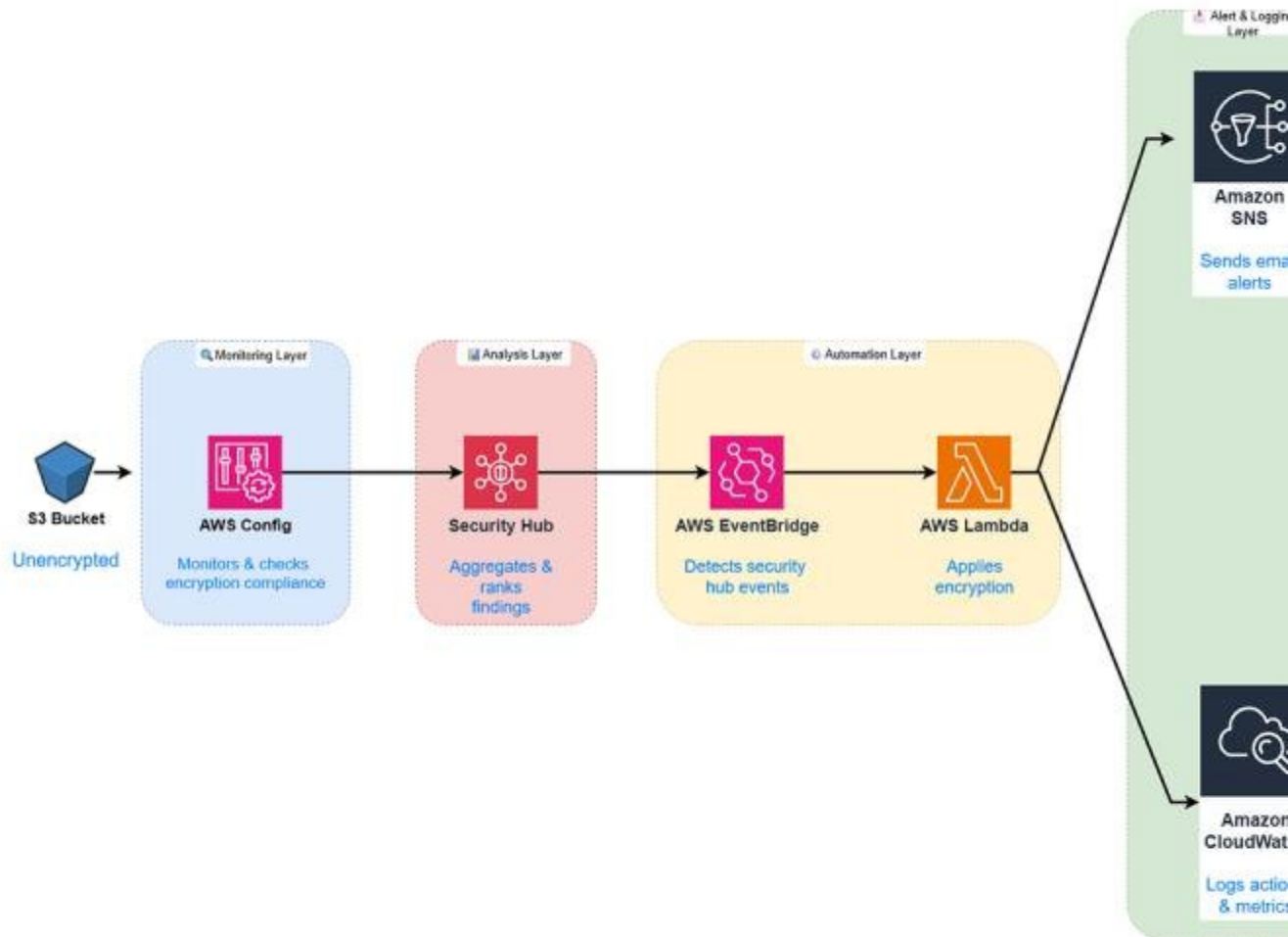
EU Partner Key Bank

EU Region



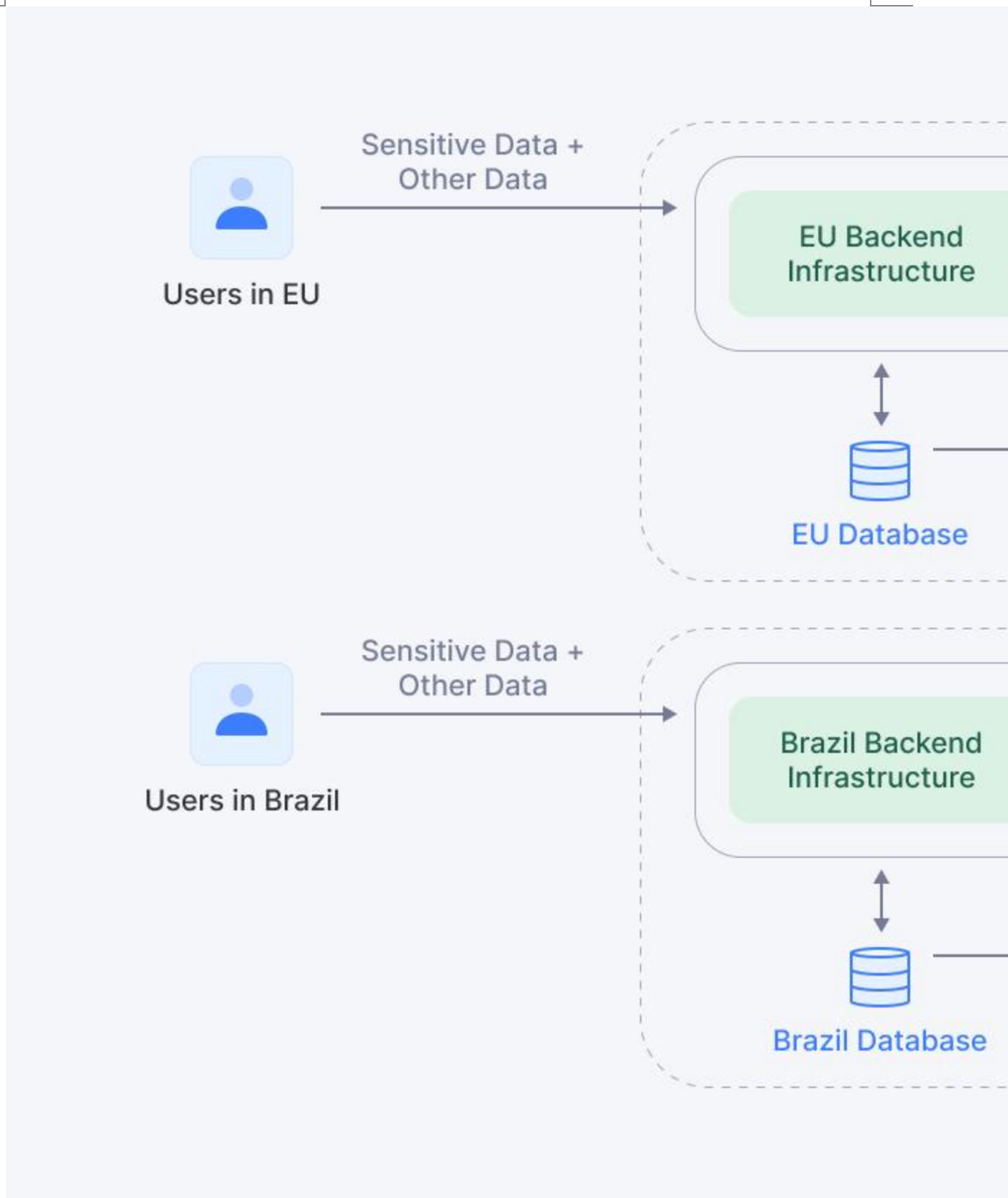
Secure Cloud Architecture Des





Automated Encryption Compliance System on AWS





Introduction

Financial institutions handle extremely sensitive information such as:

- Bank account details
- Credit/debit card information
- Customer identity data
- Payment transactions
- Loan information
- Financial statements

Therefore, cloud infrastructure must provide strong security, auditing, monitoring, encryption, and compliance controls.

1. Regulatory Compliance

Cloud providers support organizations with compliance-related controls and certifications.

Examples of commonly relevant frameworks/regulations include:

- PCI DSS
- ISO 27001
- SOC reports
- GDPR, where applicable
- Regional financial regulations

AWS, for example, provides compliance programs and certifications covering multiple industries and geographic requirements. ([AWS Documentation](#))

However, compliance is a shared responsibility: the provider secures the underlying cloud infrastructure, while the customer must configure its applications, identities, data, and policies correctly.

2. Logging

Logging records important activities.

Example:

User Login

!

Transaction Created

!

Payment Approved

!

Database Updated

!

Audit Log Stored

Logs can record:

- User login
- Failed login
- Transaction
- Data access
- Configuration changes
- Administrator actions
- Permission changes

This creates an audit trail for investigation and compliance.

3. Monitoring

Monitoring continuously checks the environment for suspicious or abnormal behavior.

For example:

50 failed logins from one account Alert security team.

Monitoring can detect:

- Unusual login patterns
- High network traffic

- Unauthorized access
- Privilege escalation
- Failed transactions
- Configuration changes

4. Encryption

Encryption protects financial data from unauthorized disclosure.

Customer Data

Encryption

Encrypted Cloud Storage

Two important cases are:

Data at rest: Encrypt databases, backups, and storage.

Data in transit: Use TLS/HTTPS when data travels between users, applications, and cloud services.

Cloud platforms provide encryption and access-control mechanisms to protect customer data. ([Amazon Web Services, Inc.](#))

5. Data Sovereignty

Data sovereignty means data is subject to the laws of the jurisdiction where it is located.

A financial organization can address this by:

1. Selecting an appropriate cloud region.
2. Restricting where databases can be created.

3. Preventing unauthorized cross-region replication.
4. Applying IAM policies.
5. Encrypting sensitive information.
6. Monitoring data movement.
7. Maintaining audit records.

Cloud providers offer controls for selecting data locations and enforcing residency-related policies. ([Amazon Web Services, Inc.](#))

Financial Transaction Example

Consider an online banking transaction:

Customer

Mobile Banking App

Encrypted Connection

Cloud API

Transaction Service

Fraud Detection

Database

Audit Log

Suppose a customer transfers ₹50,000.

The system:

1. Authenticates the customer.
2. Encrypts the communication.
3. Validates the transaction.
4. Checks fraud/security rules.
5. Updates the account database.
6. Records the transaction in audit logs.
7. Generates security alerts if abnormal behavior occurs.

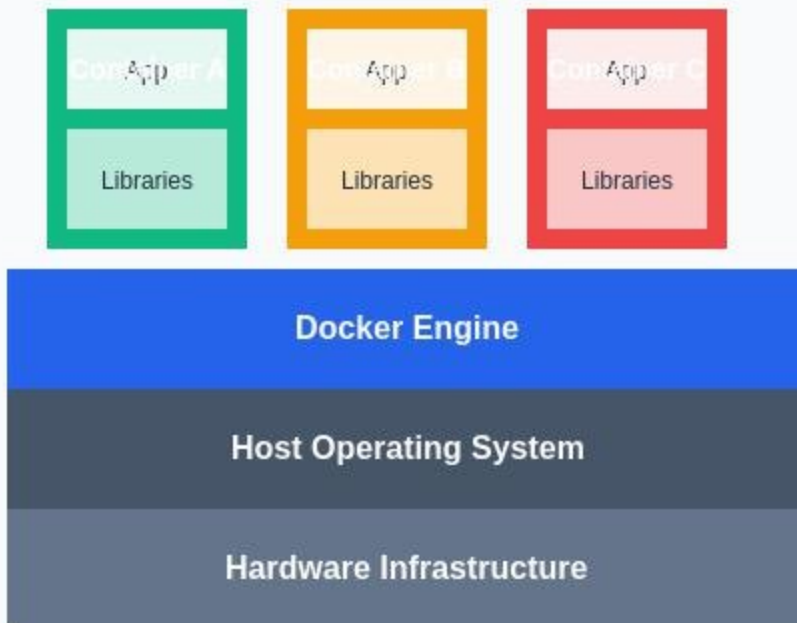
Conclusion: Cloud security for financial institutions depends on compliance controls, encryption, IAM, logging, monitoring, auditing, and data-residency controls. Together they protect sensitive financial transactions and support regulatory requirements.

5. Containerization, Kubernetes and Microservices

Container vs Virtual Machine Architecture

Docker Containers

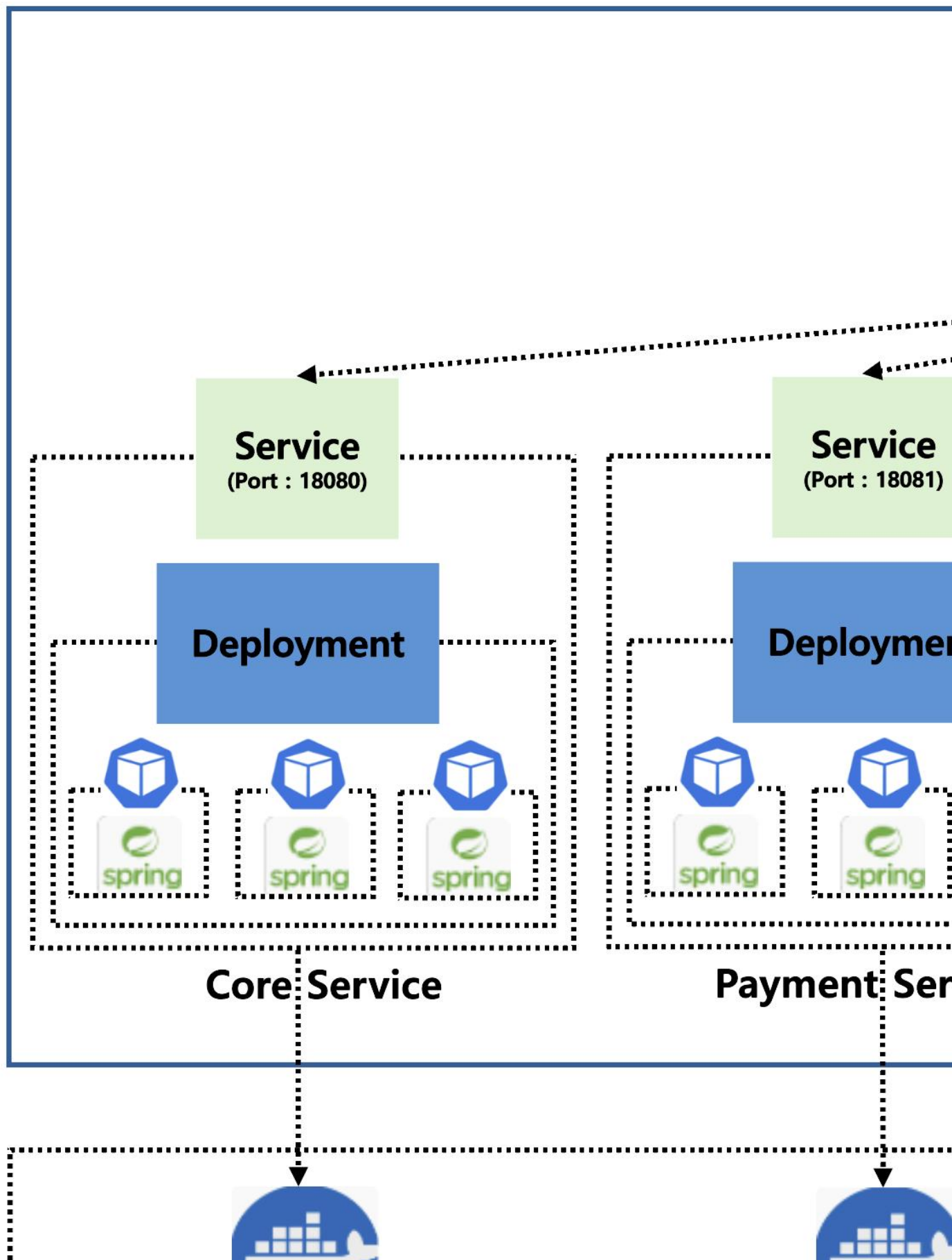
- ✓ Shared OS Kernel
- ✓ Lightweight
- ✓ Fast Startup

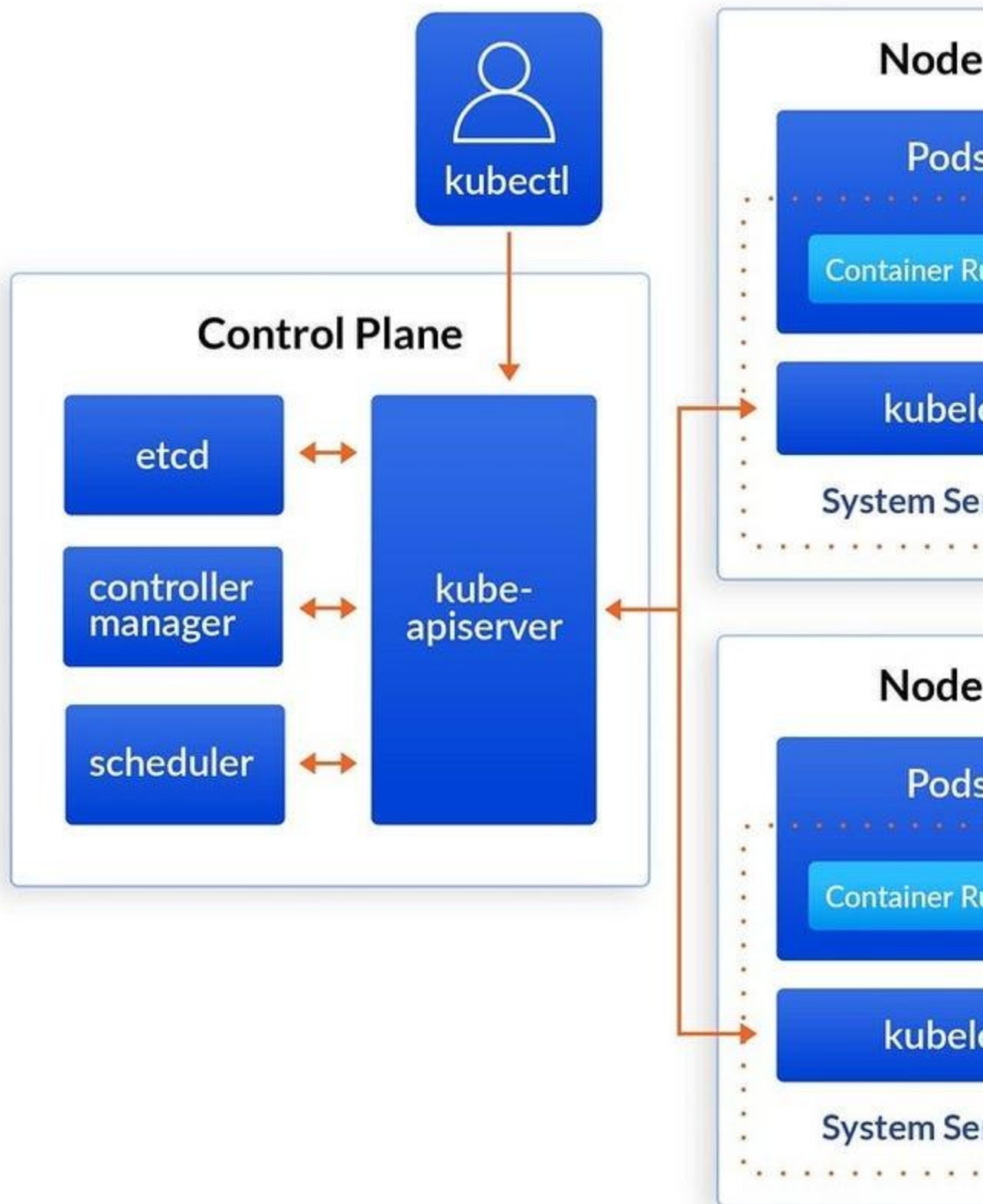


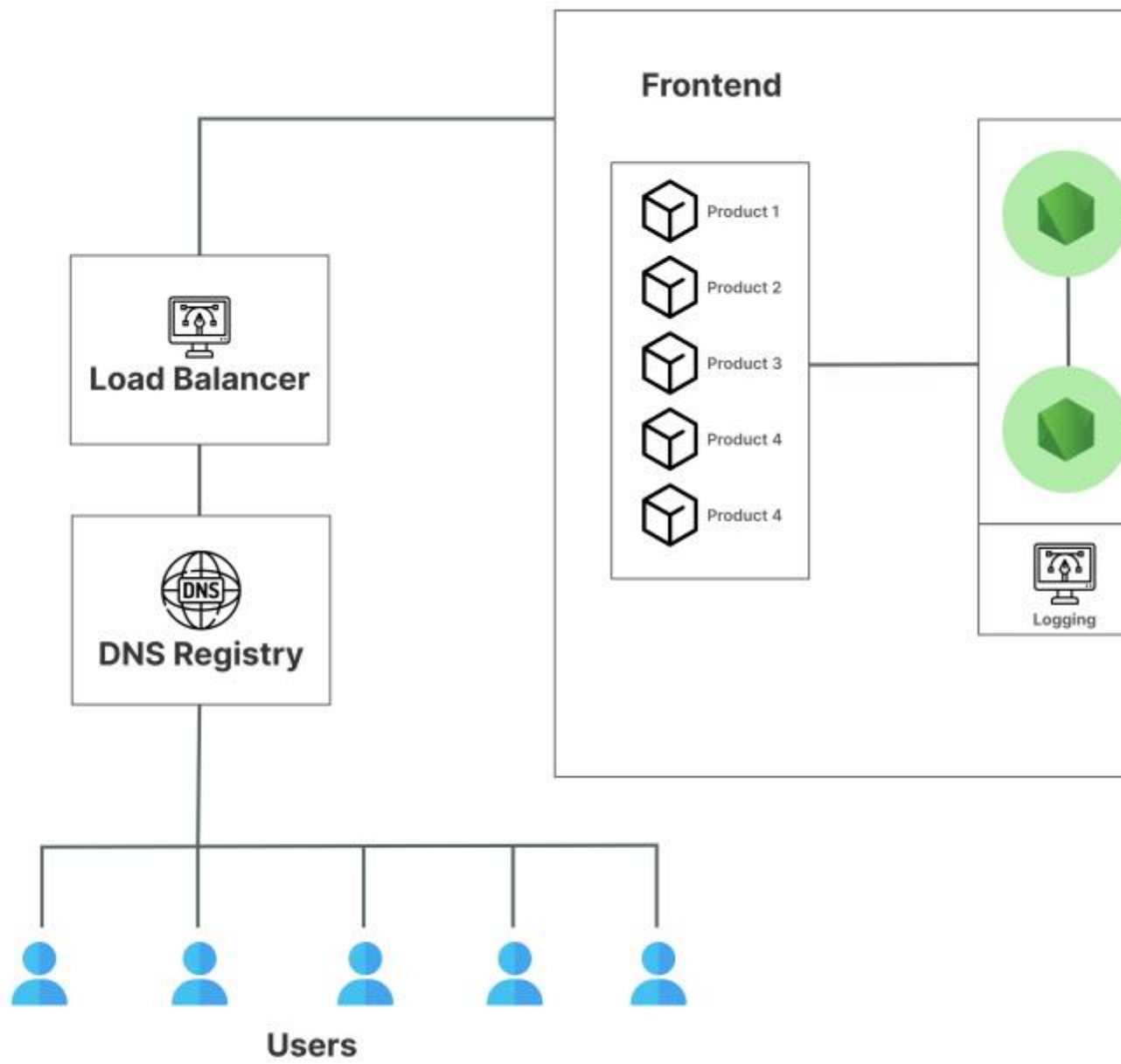
VMware

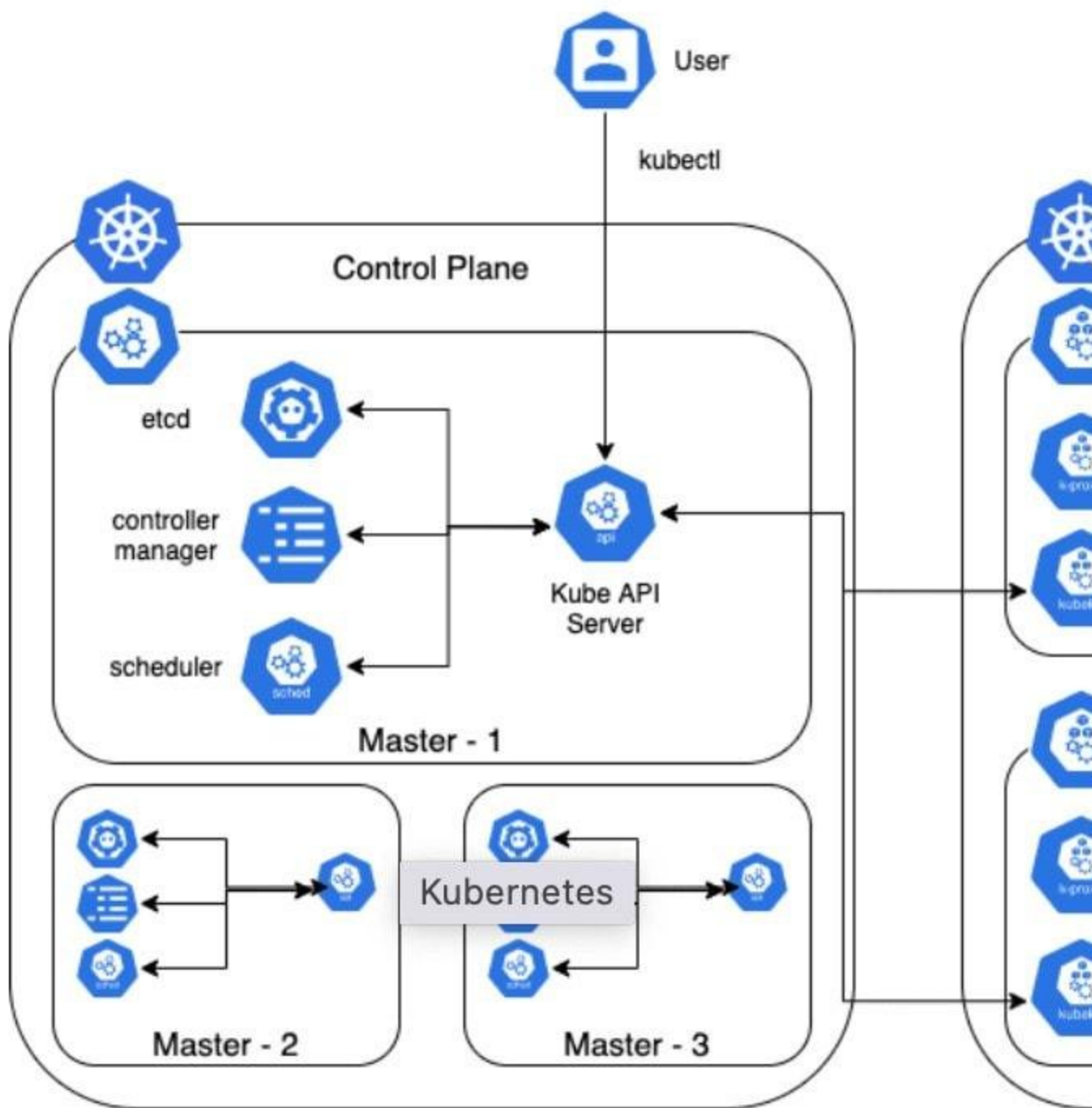
- Complete Isolation
- Higher Overhead
- Slower Startup











Introduction

Containerization packages an application together with its dependencies so that it can run consistently across different environments.

A popular container platform is Docker, while Kubernetes is widely used for container

orchestration.

Containers vs Virtual Machines

Virtual Machine Architecture

Hardware

Hypervisor

VM 1 VM 2 VM 3

OS OS OS

App App App

Each VM generally includes its own guest operating system.

Container Architecture

Hardware

Host Operating System

Container Runtime

App 1 App 2 App 3

Container Container Container

Containers share the host OS kernel, making them generally lighter and faster to start than full VMs. Kubernetes documentation describes containers as lightweight because they share the operating system, while VMs include a full machine/guest OS. ([Kubernetes](#))

Comparison

Feature	Virtual Machine	Container
OS	Separate guest OS	Shares host OS kernel
Size	Usually larger	Usually smaller
Startup	Slower	Faster
Resource usage	Higher	Lower
Isolation	Stronger hardware/OS boundary	Process-level isolation
Portability	Good	Very high
Density	Lower	Higher
Deployment	Relatively slower	Rapid

Role of Kubernetes

Managing hundreds of containers manually would be difficult.

Kubernetes automates container management.

Kubernetes Architecture


```
graph TD; Users[USERS] --> LB[Load Balancer]; LB --> K8s[Kubernetes Cluster]; K8s --> P1[Pod 1 Frontend]; K8s --> P2[Pod 2 Payment]; K8s --> P3[Pod 3 Orders]; P1 --> DB[(Database)]; P2 --> DB; P3 --> DB;
```

USERS

Load Balancer

Kubernetes Cluster

Pod 1	Pod 2	Pod 3
Frontend	Payment	Orders

Database

Kubernetes can help with:

- Container deployment
- Scaling
- Load distribution
- Service discovery
- Health checking
- Self-healing
- Rolling updates
- Rollbacks

- Resource management

Automatic Scaling

Suppose an application normally uses:

3 containers

During a traffic spike:

3 → 10 containers

After traffic decreases:

10 → 3 containers

Kubernetes can manage this scaling based on configured policies and workload requirements.

Portability Across Cloud Environments

A major advantage of containers is that the same container image can be used in different environments.

For example:

Developer Laptop

!

Docker Container

!

Testing Environment

!

AWS Kubernetes

!

Azure Kubernetes

!

Google Cloud Kubernetes

The application and its dependencies are packaged together, reducing environment-related inconsistencies. Kubernetes documentation highlights container portability across operating systems and major cloud environments. ([Kubernetes](#))

Advantages

Feature	Benefit
Containers	Lightweight deployment
Kubernetes	Automated management
Auto-scaling	Handles changing workloads
Self-healing	Restarts failed containers
Rolling updates	Reduced deployment downtime
Portability	Works across cloud environments
Microservices	Independent development and scaling
Resource efficiency	Higher application density

Conclusion: Containers provide lightweight and portable application packaging, while Kubernetes automates deployment, scaling, networking, and recovery. Together, they are highly suitable for modern microservices-based cloud applications.